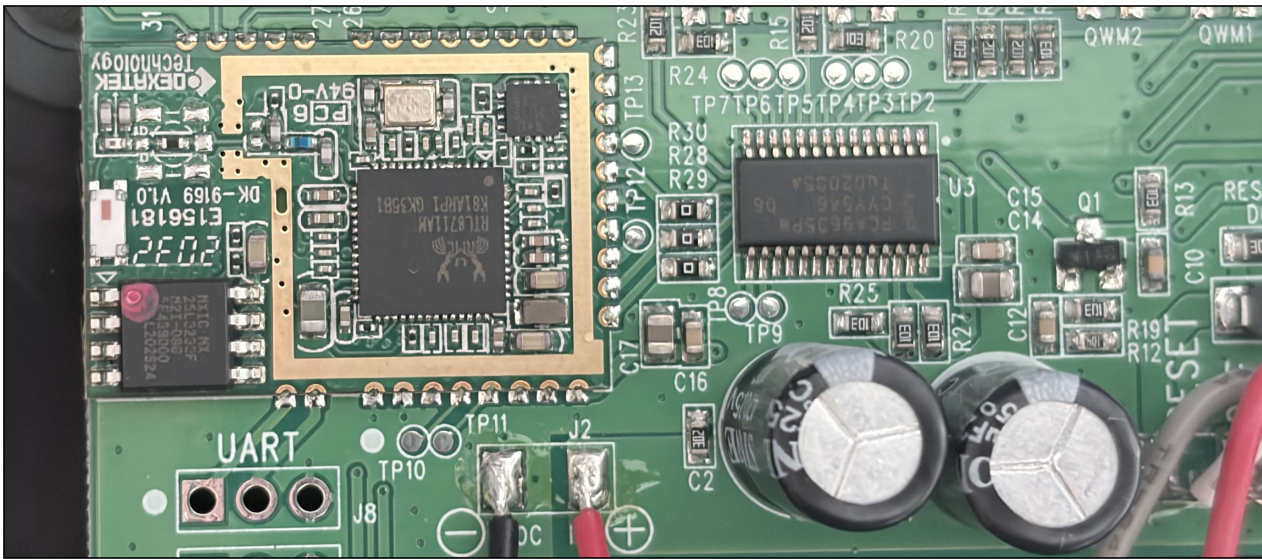


Find and circle the pins

Confirm each connection with a multimeter, then circle its pad on the photo and map.



CIRCLE

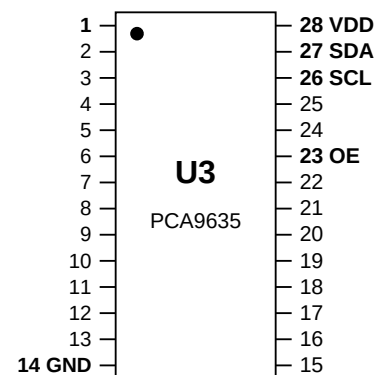
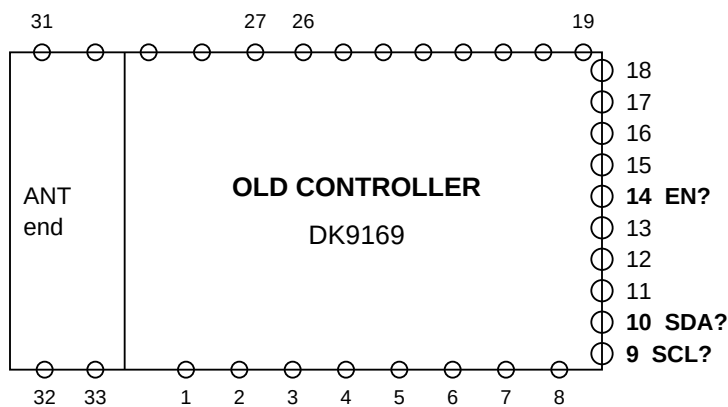
SDA

SCL

EN

J2 +

J2 -



Circle confirmed pads. The ? labels are candidates; check them with the meter.

Top view. Match the dot/notch on the real chip before counting pins.

1 Open the lamp

Unplug it. Circle **J2 + and -** in the photo; these are the power input pads. Find the Realtek module and U3.

2 Find U3 pin 1

Match its dot/notch to the drawing. Circle pins **27 SDA, 26 SCL and 23 OE** on the map.

3 Find SDA and SCL pads

With power off, check continuity: module **10 to U3 27**, and **9 to U3 26**. Circle the confirmed module pads.

4 Find the enable wire

Trace **U3 23 (OE)** to the module; try **pad 14**. Circle that pad as EN. If OE is tied to GND, write **NO EN WIRE**.

5 Check the voltages

Power the stock board. J2 + to -: ____ V (about **13 V**). SDA / SCL to J2 -: ____ / ____ V (about **3.3 V**). Unplug.

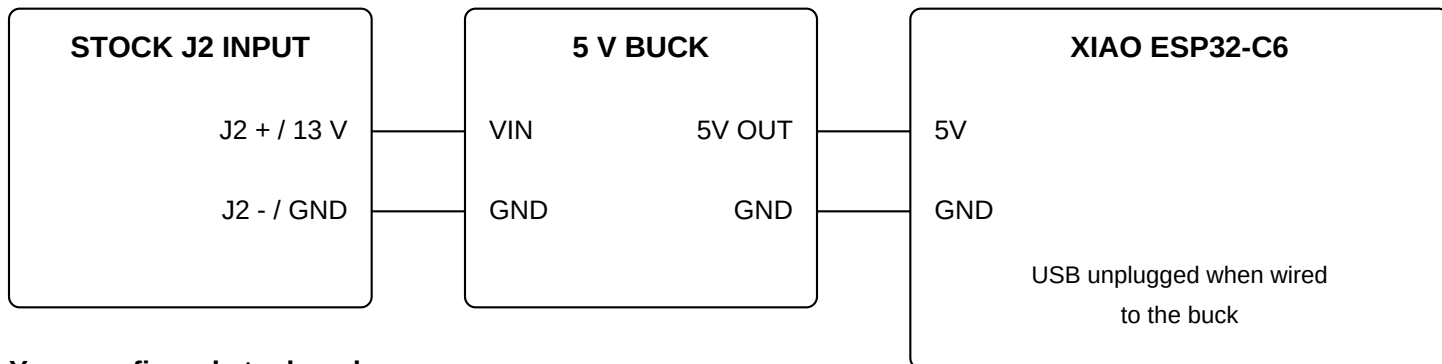
6 Remove the controller

Desolder only the Realtek module. Keep U3 and its pins intact. Use the pads you circled for the new wires.

If a connection is unclear or the signal voltage is not about 3.3 V, stop and record it before wiring the ESP.

Wire it, pair it, try it

Parts: XIAO ESP32-C6 + Pololu D24V25F5 (5 V buck), wire and insulated mounting. Keep the stock LED circuitry.



Your confirmed stock pads

SDA: _____ SCL: _____ EN: _____

SDA	_____→	D10 / GPIO18
SCL	_____→	D9 / GPIO20
EN / OE*	_____→	D3 / GPIO21

*Connect EN only to the controller pad you traced to OE. If OE is tied to GND, leave D3 unconnected.

7 Flash before wiring

Connect only the ESP to USB. From the repo, list its port, flash, then read its pairing code:

```
python3 scripts/device.py --list
sh scripts/flash.sh /dev/cu.usbmodemPORT
python3 scripts/device.py --port /dev/cu.usbmodemPORT commissioning code
```

Replace the port above. Pairing code: _____ Then unplug USB.

8 Connect power

Wire the buck to J2 + and -. Power the lamp and check for **5 V** at the buck output. Unplug, then connect the ESP.

9 Connect the control wires

Solder SDA, SCL and the identified EN wire as drawn. Reconnect any unplugged LED leads to their original sockets.

10 Pair in Apple Home

Power the lamp. In Home, add an accessory using the code above. Pair within **15 minutes**; power-cycle to reopen the window.

11 Try both presets

Try both presets and Off. Rename them 3300 K (warm) and 5000 K (cool). Turn one on; power-cycle and check it returns.

12 Mount and close

Unplug. Insulate and secure the ESP and buck, then close the lamp. Check Home still reaches it.